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EXTRA



SOMAIYA
VIDYAVIHAR UNIVERSITY

19.11.2024 (E)

Maximum Marks: 100	Semester: July 2024 –November 2024	Duration: 3 Hrs.
Programme code: 03	Examination: ESE Examination	
Programme: Electronics and Telecommunication Engineering	Class: TY BTECH	Semester: V(SVU 2020)
Name of the Constituent College: K. J. Somaiya School of Engineering	Name of the department: Electronics and Telecommunication Engineering	
Course Code: 116U03C501	Name of the Course: Digital Communication	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	
i)	State the advantages and disadvantages of digital communication	20
ii)	What is significance of uncertainty in information source?	5
iii)	State sampling theorem. Calculate Nyquist rate for voice signal of range 300 Hz to 3400 Hz	5
iv)	What is the significance of signal space diagram?	5
v)	A FM signal at 10.7 MHz IF needs to be digitized for demodulation in a digital domain. If the bandwidth of the signal is 200 kHz, find the minimum usable sampling frequency	5
vi)	Explain the effects of noise on the channel. Also state the need of channel modelling.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	
i)	For two messages, show that entropy is minimum when the messages are extremely unlikely or extremely likely and maximum when equally likely. Also plot it as a function of probability of one of the messages	10
ii)	Find capacity of Gaussian channel of bandwidth 4KHz and noise PSD 10^{-9} W/Hz when signal energy is a) 0.1J b) 0.001J c) how does channel capacity changes in b) if bandwidth is increased to 10KHz?	5
OR		
Q2 A	Consider the four signals $s_1(t)$, $s_2(t)$, $s_3(t)$ and $s_4(t)$ as shown in the fig-P1.1. Use Gram-Schmidt Orthogonalization Procedure to find the orthonormal basis for this set of signals. Also express the signals in terms of the basis functions	10

Q 2 B	Solve any One	10
i)	Using a block diagram explain the basic Signal Processing Operations in Digital Communications	10
ii)	Determine the Lempel Ziv code for the following stream 1010110110101011 Recover the original sequence from the encoded stream	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	For a (7,4) block code, the received code word is [1 1 1 0 0 0 0]. Is this code word correct? If not, then correct it using syndrome decoding. Parity matrix is given as: $\begin{matrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{matrix}$	10
ii)	The generator polynomial for a (7,4) cyclic code is $g(x) = 1 + x^2 + x^3$ Find the code vector (in systematic form) for the message polynomials $D_1(x) = x^3 + x$, $D_2(x) = x^3$	10
iii)	Draw a systematic convolutional encoder of rate $\frac{1}{2}$, constraint length 4 and explain its working.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	MSK is called "Shaped QPSK". Justify with relevant expressions & waveforms. Discuss the merits and demerits of MSK as compared to QPSK	10
ii)	Explain the ASK system and derive the relation for error probability of binary ASK	10
iii)	Compare MPSK and MFSK systems	10

Que. No.	Question	Max. Marks
Q5	Write Short notes on any four	20
i)	Hartley Shannons theorem	5
ii)	Channel coding theorem	5
iii)	Geometric representation of signals	5
iv)	Viterbi algorithm	5
v)	Spectrum of flat top sampled signal.	5
vi)	Gaussian MSK	5

21.11.2024 (E)


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Semester: July 2024 –November 2024

Maximum Marks: 100

Examination: ESE Examination

Duration: 3 Hrs.

Programme code: 03

Programme: B.Tech EXTC

Class: TY

Semester: V(SVU 2020)

Institute/School/Department: K. J. Somaiya School of Engineering

Name of the department: EXTC

Course Code: 116U03C502

Name of the Course: RF Filters and Antenna

 Instructions: 1) Draw neat diagrams 2) All questions are compulsory
 3) Assume suitable data wherever necessary

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Explain high frequency equivalent circuit of inductor with diagram, high frequency response, and equation.	5
ii)	Explain Gain and Directivity of an antenna and relationship between them.	5
iii)	Draw well labelled frequency response of band stop filter and explain the significance of filter parameters.	5
iv)	Explain quarter wave transmission line and its use in impedance matching.	5
v)	Explain significance of primary element, reflectors and directors in the Yagi-Uda antenna.	5
vi)	Write short note on pattern multiplication using 4 isotropic point sources fed with equal amplitude, equal phase and separated by a distance of $\lambda/2$ from each other.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	The radiation resistance of an antenna is 52 ohms and loss resistance is 8 ohms. What is the directivity if power gain is 16.	5
ii)	Explain current distribution on $\lambda/2$ dipole and λ dipole.	5
	OR	
Q2 A	Draw the lumped circuit model for a transmission line. Derive expressions for voltage and current travelling waves.	10
Q 2 B	Solve any One	10
i)	A transmission line of characteristic impedance 50Ω and length $d = 0.15\lambda$ is terminated in load impedance of $Z_L = (30 - j40) \Omega$. Find τ_0 , $Z_{in}(d)$ and SWR by using Z- smith chart.	10
ii)	Design a low pass butterworth filter that will provide atleast 20dB attenuation at $f = 2f_{3dB}$ and source and load resistance of 50Ω . Refer below table of prototype values.	10

Coefficients for maximally flat low-pass filter ($N = 1$ to 10)

N	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	g_9	g_{10}	g_{11}
1	2.0000	1.0000									
2	1.4142	1.4142	1.0000								
3	1.0000	2.0000	1.0000	1.0000							
4	0.7654	1.8478	1.8478	0.7654	1.0000						
5	0.6180	1.6180	2.0000	1.6180	0.6180	1.0000					
6	0.5176	1.4142	1.9318	1.9318	1.4142	0.5176	1.0000				
7	0.4450	1.2470	1.8019	2.0000	1.8019	1.2470	0.4450	1.0000			
8	0.3902	1.1111	1.6629	1.9615	1.9615	1.6629	1.1111	0.3902	1.0000		
9	0.3473	1.0000	1.5321	1.8794	2.0000	1.8794	1.5321	1.0000	0.3473	1.0000	
10	0.3129	0.9080	1.4142	1.7820	1.9754	1.9754	1.7820	1.4142	0.9080	0.3129	1.0000

Que. No.	Question	Max. Marks
Q3	Solve any Two	
i)	State importance of FRISS Equation and solve the following problem. In microwave communication link, two identical antennas operating at 10 GHz are used with power gain of 40dB. If the transmitter power is 3 W, find the received power, if the range of the link is 30 km.	20 10
ii)	Discuss the structure of microstrip antenna with the help of transmission line model and contact type feeding techniques in microstrip antennas.	10
iii)	1. Determine HPBW and FNBW, if radiation intensity is $E(\theta) = \cos \theta$ where ($0 \leq \theta \leq 90^\circ$, $0 \leq \Phi \leq 360^\circ$) 2. Explain fresnel and fraunhofer regions of an antenna.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	
i)	Derive the E field for a. Array of two isotropic point sources of same amplitude and same phase b. Array of two isotropic point sources of same amplitude and out of phase.	20 10
ii)	A broadside array consists of 4 isotropic point sources with a distance $\lambda/2$ between them. Find i. array factor ii. Direction of major and minor lobes. iii. Direction of minima iv. HPBW and FNBW. v. Plot the radiation pattern.	10
iii)	Discuss cassegrain feeding technique in reflector antenna	10

Que. No.	Question	Max. Marks
Q5	Solve any four (Short notes / Short question type)	
i)	A magnetic field strength of $15 \mu\text{A/m}$ is required at point on $\theta = \pi/2$, 2 km from an antenna in air. Neglecting ohmic loss, how much power must the antenna transmit if it is an infinitesimal dipole of length of $\lambda/50$.	20 5

ii)	A lossless transmission line 80 cm has $L = 0.25\mu\text{H/m}$ and $C=100\text{pF/m}$, find the characteristic impedance of line.	5
iii)	Explain construction and working of helical antenna.	5
iv)	Explain different types of radiation pattern of an antenna.	5
v)	Plot the following impedance points and their reflection coefficient and VSWR on the impedance smith chart .Given $Z_0 = 50 \Omega$. $Z_1 = (23 + j 42) \Omega$, $Z_2 = (115 - j 22) \Omega$	5
vi)	Explain skin depth with formula.	5

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23.11.2024 (E)

Semester: July 2024 –November 2024			
Maximum Marks: 100		Examination: ESE Examination	
Programme code: 03		Duration:3 Hrs.	
Programme: B.Tech		Class:TY	Semester: V (SVU 2020)
Institute: K. J. Somaiya School of Engineering		Name of the department: EXTC	
Course Code: 116U03E513	Name of the Course: Operating Systems		
Instructions: 1)Draw neat diagrams 2) All questions are compulsory			
3) Assume suitable data wherever necessary			

Que. No.	Question	Max. Marks
Q1	Solve any Four	
i)	Describe the process of booting an OS.	20
ii)	Explain files in relation with File access methods.	5
iii)	Differentiate between user level thread and kernel level thread.	5
iv)	What is demand paging, and how does it differ from pre-paging?	5
v)	What is a system call? List some common system calls.	5
vi)	Define the term Operating System (OS), and what are its primary functions?	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	
i)	Describe the key characteristics of a Real-Time Operating System.	10
ii)	How does an RTOS handle multitasking compared to a general-purpose OS?	5
	OR	5
Q2 A	What are interrupts, and how do they work in an OS?	10
Q 2 B	Solve any One	
i)	What is a kernel? Differentiate between monolithic and microkernel.	10
ii)	Explain clearly paging and segmentation based memory management techniques using diagram.	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	
i)	Suppose that a disk drive has 200 cylinders, numbered 0 to 199. The initial head position is at 53 th cylinder. The queue of pending requests in FIFO is 99, 184, 38, 121, 15, 125, 68 and 71. Calculate average seek time for each of the following algorithm. (Assume: 15ms required to move 1 cylinder) Given:- Previous Arm request=25 1. FCFS 2. SSTF 3. SCAN 4. C-SCAN	20
ii)	Explain Rate monotonic Algorithm (RMA) & also give important criteria (necessary and sufficient conditions) to check schedulability of a task set under RMA.	10
iii)	Differentiate between contiguous, linked, and indexed file allocation methods:	10

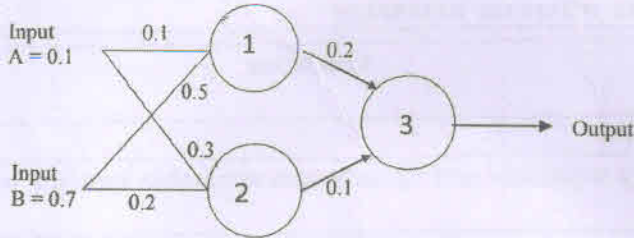
Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	What is disk Arm Scheduling? What are the various disk scheduling algorithms? Explain the criteria for selecting the best scheduling algorithm.	10
ii)	What is RAID, explain any 5 RAID levels.	10
iii)	Explain the types of file directory system.	10

Que. No.	Question	Max. Marks
Q5	Solve any four	20
i)	Write a note on virtual memory.	5
ii)	Define mutual exclusion. What are some methods to achieve it?	5
iii)	Explain what a page table is and its role in memory management.	5
iv)	What is a critical section, and how is it managed?	5
v)	Compare pre-emptive and non pre-emptive scheduling algorithm	5
vi)	Write a note on UNIX goals.	5



Semester: July 2024 –October 2024		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code:03		
Programme: B. Tech., Electronics and Telecommunication Engineering	Class: T.Y.	Semester: V(SVU 2020)
Name of the Constituent College:		
K. J. Somaiya College of Engineering	Name of the department: EXTC	
Course Code: 116U03E514	Name of the Course: Soft Computing	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Explain linearly separable and linearly non-separable systems with suitable example.	5
ii)	State the hyperparameters that affect learning in Multiple Layer Perceptron networks.	5
iii)	What is fuzziness? Explain with example.	5
iv)	Compare sigmoid and ReLU activation functions.	5
v)	What is the purpose of fitness function evaluation in genetic algorithm?	5
vi)	The fuzzy set A is defined on universe, $x = [0,1,2,3]$ with the following membership function: $\mu_A(x) = \frac{2}{x+3}$ Define the intervals along the x-axis corresponding to the α – cut set for fuzzy set A for $\alpha = 0.2$ and $\alpha = 0.45$.	5
Q2 A	Solve the following	10
i)	State what can be the reason for, each of the following observations, when a neural network is trained on the train data and evaluated on validation data:- 1. Training error : low; Validation error: low 2. Training error : low; Validation error: high 3. Training error : high; Validation error: high 4. Training error : high; Validation error: low	5
ii)	Explain the function of following layers in 2-Dimensional Convolutional Neural Network employed for image classification: A. Convolution layer B. Pooling layer C. Dropout layer D. Fully connected layer E. Output layer	5
OR		
Q2 A	Describe architecture of Radial Basis Function neural network with the help of an appropriate sketch. What are the parameters of this network? Explain the learning process to learn these parameters.	10

Que. No.	Question	Max. Marks
Q 2 B	Solve any One	10
i)	Consider a MLP network shown below. Assume sigmoid activation function for neurons in hidden and output layer. Assume learning rate of 1 and target of 1 for the given inputs A and B. - Apply the forward pass and find output of nodes 1, 2 and 3 - Compute error at output layer neuron - Compute updated weights of the connections between hidden-output layers - Compute error at hidden layer neurons - Compute updated weights of the connections between input- hidden layers 	
ii)	Using Kohonen SOM, RGB color data, where $x_1=R$, $x_2=G$, $x_3=B$, is used to classify chocolate images in two clusters/classes: brown or white. Initial weight matrix is: $[(0.2, 0.6, 0.5), (0.8, 0.4, 0.7)]$, learning rate is 0.5. - Draw the architecture of SOM - For a particular RGB value (1.0, 0.5, 0.14) of a chocolate image, find BMU based on Euclidean distance calculations. - Write the updated weight matrix.	
Q3	Solve any Two	20
i)	Consider the following real variables from everyday life: - Income measured in Rs. - Speed measured in meters per second. - A TV show measured in how much you are interested watching it. - A meal measured in how much you like to eat it. - A traffic light measured in what colour is on. In each case, suggest a fuzzy variable corresponding to these real variables. For which of these five variables the use of a fuzzy variable is not really necessary? Why?	10
ii)	Describe Mamdani Fuzzy Inference system for controlling a room temperature. Assume relevant input and output variables.	10
iii)	Classify the methods of defuzzification? Explain Center of Gravity method in detail.	10
Q4	Solve any Two	20
i)	Describe the basic Genetic Algorithm operators along with their purpose.	10
ii)	Describe the various Encoding operators in Genetic algorithm.	10
iii)	Let the initial population be 0 1 1 0 0, 1 1 0 0 1, 0 0 1 0 1, 1 0 0 1 1. Maximize the function $f(x) = x^2$; where x is permitted to vary between 0 and 31.	10

Que. No.	Question	Max. Marks
Q5	Answer any four	20
i)	Describe Transfer learning and state its advantages.	5
ii)	How does a momentum factor in gradient descent algorithm help for faster convergence of neural network learning algorithm?	5
iii)	A MLP neural network architecture has: 4 input nodes, single hidden layer with 5 hidden nodes, and output layer having 3 output nodes. It will require X number of weights and Y number of biases to be trained. Write values of X & Y.	5
iv)	Fuzzy sets A and B are given below: $A = \{ 0/0 + 0.2/1 + 0.7/2 + 0.8/3 + 0.9/4 + 1/5 \}$ $B = \{ 0/0 + 0.1/1 + 0.3/2 + 0.2/3 + 0.4/4 + 0.5/5 \}$ Express for A and B: (a) Algebraic sum (b) Algebraic product (c) Bounded sum (d) Bounded difference (e) Complement of A	5
v)	Differentiate between Roulette wheel selection and tournament selection.	5
vi)	Explain in brief architecture of AlexNet network.	5



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Semester: July 2024 –November 2024		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 03	Class: TY	Semester: V(SVU 2020)
Programme: B.Tech Electronics and Telecommunication Engineering		
Institute/School/Department: K. J. Somaiya School of Engineering		Name of the department: EXTC
Course Code: 116U03E515	Name of the Course: Principles of Data Science	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory		
3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	A researcher is studying the heights (in cm) and genders (male, female) of students in a class over a semester. 1. Define the population and sample for this study. 2. Classify the variables height and gender into the appropriate data variable types: qualitative (nominal, ordinal, binary) and quantitative (continuous, discrete). 3. Is the dataset described a cross-sectional or time series dataset? 4. Determine whether the data is uni-variate, bi-variate, or multi-variate. 5. Suppose the researcher records the height and gender of 50 students, calculate the average height if the total height recorded is 8100 cm.	5
ii)	A class of 20 students took a math test, and their scores are as follows: 55,67,58,72,90,88,76,64,95,82,73,69,84,91,75,60,85,78,66,70 1. Construct a frequency table for the test scores 2. Draw a histogram based on the frequency table. 3. Draw ogive plot	5
iii)	Calculate the sample correlation coefficient for two variables, $X = [3, 6, 9, 12, 15]$ and $Y = [8, 10, 12, 14, 16]$, and interpret the result.	5
iv)	Explain Chebyshev's Inequality and its significance in data science. Provide an example to illustrate its application.	5
v)	For a dataset $[20, 22, 24, 26, 28, 30]$, calculate the five-number summary (minimum, Q1, median, Q3, maximum) and draw a box plot.	5
vi)	Calculate the mean, median, mode, sample variance, and standard deviation for the data: $[12, 15, 14, 19, 15, 20, 18, 16]$. Summarize what these measures tell about the data distribution.	5

Que No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	For a normally distributed population with sample mean 50 and population variance 16, find the 95% confidence interval for the mean, assuming a sample size of 30.	5
ii)	For a discrete random variable X with values [1, 2, 3, 4, 5] and corresponding probabilities [0.1, 0.2, 0.3, 0.2, 0.2], calculate the expected value of X and expected value of X^2 .	5
	OR	
Q2 A	A factory produces light bulbs, and the lifetime of the bulbs is normally distributed with a mean μ of 800 hours and a standard deviation σ of 50 hours. A quality control manager takes a random sample of 64 bulbs. 1. Calculate the standard error of the mean. 2. Determine the probability that the sample mean lifetime of the 64 bulbs is more than 810 hours. 3. Interpret the result in the context of the Central Limit Theorem.	10
Q 2 B	Solve any One	10
i)	Derive Maximum Likelihood Estimate (MLE) for a Bernoulli distribution. Calculate the MLE for the mean of a Bernoulli distribution with observations [0, 1, 1, 0, 1, 1]	10
ii)	A researcher wants to estimate the variance of the weights of a certain species of birds. The researcher collects a random sample of 20 birds and records their weights (in grams). The sample variance is found to be 25. 1. Construct a 95% two sided, lower and upper confidence interval for the population variance. 2. Provide an interpretation of the confidence interval in the context of the study.	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	A manufacturing company claims that the average diameter of its ball bearings is 5 cm with a known standard deviation of 0.2 cm. A random sample of 25 ball bearings is selected, and the sample mean diameter is found to be 4.95 cm. At the 5% significance level, test whether the mean diameter of the ball bearings is significantly different from 5 cm using a Z-test.	10
ii)	A study is conducted to determine if a new teaching method improves student test scores. A random sample of 16 students taught with the new method has a mean score of 78 with a standard deviation of 10. The average score of students taught with the traditional method is known to be 75. Test at the 5% significance level whether the new teaching method has a significantly different mean score using a T-test.	10
iii)	A machine fills cans with a standard deviation of 0.5 kg. To test the quality, 50 cans were sampled, yielding a sample standard deviation of 0.6 kg. Test if the process variance has increased at a 5% significance level.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Compare the effectiveness of three diets on weight loss by analysing data for five individuals under each diet. Diet 1: [5, 6, 5, 4, 7], Diet 2: [6, 5, 7, 6, 8], Diet 3: [7, 8, 7, 9, 6]. Conduct a one-way ANOVA at a 5% significance level.	10
ii)	Explain the concept of Analysis of Variance (ANOVA) and its importance in statistical data analysis. Include the following in your explanation: 1. The purpose of ANOVA and situations where it is used. 2. Types of ANOVA (one-way and two-way) and a brief description of each. 3. Test statistics in ANOVA (TSS, SST, ESS, DF, Fcal). 4. Interpretation of ANOVA results, focusing on understanding the F-statistic and F table value. 5. An example scenario where ANOVA would be an appropriate method for data analysis.	10
iii)	Data on the performance of three different teaching methods across four classrooms gives the following outcomes for each method: [50, 52, 51, 55], [54, 57, 53, 56], [52, 53, 54, 57]. Conduct a two-way ANOVA to assess the interaction effect for methods and classrooms at a 5% significance level.	10

Que. No.	Question	Max. Marks
Q5	Solve any four	20
i)	Given a logistic regression equation with coefficients $\beta_0 = -1.2$ and $\beta_1 = 0.5$, calculate the probability that $Y = 1$ for $X = 2$.	5
ii)	Describe the concept of the Coefficient of Determination (R^2) in linear regression and explain what it indicates about the fitted model.	5
iii)	Explain the Least square method of Linear regression	5
iv)	Compare Linear and Logistic regression	5
v)	Using the following data points for sales and advertising spend: Sales = [100, 150, 200, 250], Advertising Spend = [20, 30, 40, 50]. Compute the slope and intercept for a linear regression line.	5
vi)	Explain the differences between multiple regression and polynomial regression, and discuss the scenarios in which each type of regression model would be most appropriately applied.	5

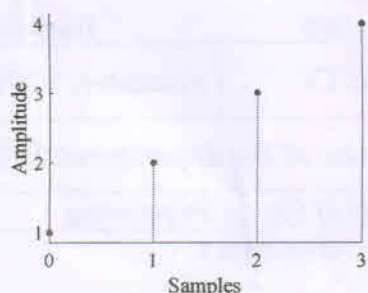


Semester: July 2024 –November 2024		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 03 Programme: B. TECH EXTC	Class: TY	Semester: V(SVU 2020)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: EXTC	
Course Code: 116U03C503	Name of the Course: Digital Signal Processing	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Find the unit impulse response for the following linear systems. a) $y(n) = 0.5 x(n) - 0.5 x(n-2)$; for $n \geq 0$, $x(-2) = 0$, $x(-1) = 0$ b) $y(n) = 0.75 y(n-1) + x(n)$; for $n \geq 0$, $y(-1) = 0$	5
ii)	Given the LTI is described by the unit impulse response $h(n) = (0.25)^n u(n)$. Determine whether this system is stable or not.	5
iii)	Find the linear convolution of given two sequences using the z transform only. $x(n) = 3\delta(n) + 2\delta(n-1)$, $h(n) = 2\delta(n) - \delta(n-1)$	5
iv)	Determine if the following FIR system is minimum phase, $h(n) = \{1, 5, 6\}$	5
v)	With suitable pole-zero plots and transfer function (first order only) describe the low pass and high pass filters.	5
vi)	A signal $x(t) = \sin(50\pi t)$ is sampled at a rate of i) 20Hz ii) 75 Hz, for each of these two cases, explain if you can recover the signal $x(t)$ from the sampled signal.	5

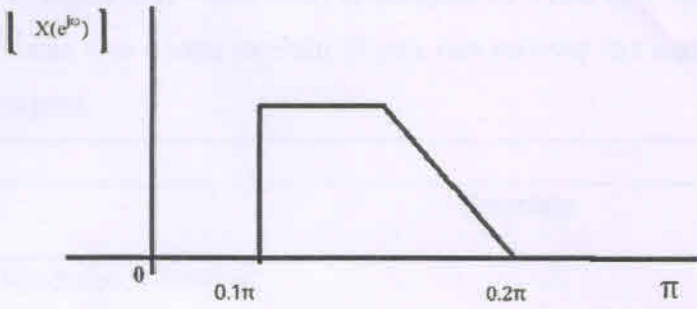
Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	DFT is used to compute the amplitude spectrum of the sampled data sequence with a sampling rate of 10 KHz. Given that it requires the frequency resolution to be less than 0.5 Hz. Determine the number of data points required by FFT algorithm assuming that data samples are available.	5
ii)	Plot the signal flow graph diagram for finding the eight point inverse RADIX-2 DIF-FFT with detailed nomenclature and directions.	5

OR

	OR	
Q2 A	Consider the following sequence and assuming sampling frequency is 100 Hz. Compute and plot the amplitude and phase spectrum of DFT of sequence. 	10
Q 2 B	Solve any One	10
i)	Given a sequence $x(n)=n$ for $0 \leq n \leq 3$, if sampling rate is 10 Hz, a) Determine the sampling period, time index and sampling time instant for a digital sample $x(3)$ in time domain. b) Determine the frequency resolution and mapped frequency for each of DFT coefficient $X(1)$ and $X(3)$ in frequency domain.	10
ii)	Perform the circular convolution of the following sequences using DFT/IDFT method. $x(n) = \{1\ 1\ 2\ 1\}$ and $h(n) = \{1\ 2\ 2\ 1\}$	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Design a digital high pass Butterworth filter with following specifications, 3dB attenuation at the pass band frequency of 1000 Hz. 10 dB stop band attenuation at the frequency of 350 Hz. - Sampling frequency of 5000 Hz. - Use BLT method.	10
ii)	Consider the following transfer Laplace transfer function $H(s) = \frac{2}{s+2}$ Determine $H(z)$ using impulse invariant method and bilinear transformation method if the sampling rate is 10 Hz. Compare the impulse invariant and bilinear transformations.	10
iii)	Design a Butterworth filter using impulse invariance method for the following specifications. $0.8 \leq [H(e^{j\omega})] \leq 1 \quad 0 \leq \omega \leq 0.2\pi$ $[H(e^{j\omega})] \leq 0.2 \quad 0.6\pi \leq \omega \leq \pi$	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Determine the transfer function for a 5-tap FIR low pass digital filter with a lower cut-off frequency of 2000 Hz and a sampling rate of 8000 Hz using the frequency sampling method.	10
ii)	Design a linear phase FIR High pass filter using Hamming window with a cut off frequency $\omega_c = 0.8 \pi$ rad/sample and $N=7$.	10
iii)	What is meant by finite word length effects in digital filters, explain with suitable examples? How it is related to quantization in a digital system.	10

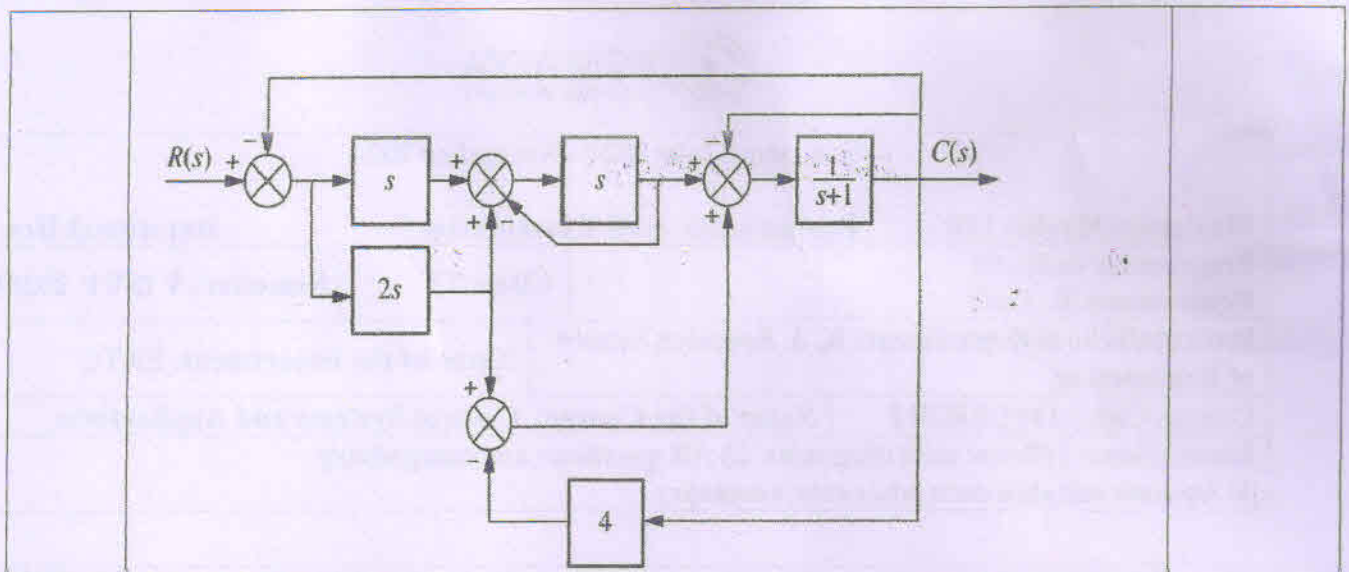
Que. No.	Question	Max. Marks
Q5	Solve any four (Short notes / Short question type)	20
i)	Describe the comb filter with its transfer function and frequency spectrum.	5
ii)	Realize the system with difference equation $y(n) = 0.75 y(n-1) - 0.125 y(n-2) + 0.5 x(n-1) + x(n)$ in cascade form .	5
iii)	A digital filter has the following impulse response $h(n) = \{-3, 2, 1, -2, 3\}$, if it is linear phase filter, justify.	5
iv)	Differentiate between FIR and IIR filters with suitable examples.	5
v)	The magnitude response of an input signal is as shown below. If input sequence is passed through the up sampler of factor 4, plot the magnitude response of the output signal. 	5
vi)	Explain the need of anti-aliasing filter prior to down sampling and anti-imaging filter after up sampling a signal.	5

23.11.2024 (E)

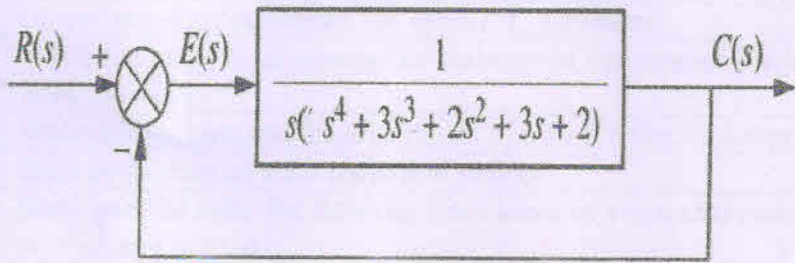

SOMAIYA
 VIDYAVIHAR UNIVERSITY

Semester: July 2024 –November 2024		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 03	Class: TY	Semester: V (SVU 2020)
Programme: B. Tech		
Institute/School/Department: K. J. Somaiya School of Engineering		Name of the department: EXTC
Course Code: 116U03E512	Name of the Course: Control Systems and Applications	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory		
3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Define state, state variables and state equations	5
ii)	Derive the error constants for Type 0, 1, 2 systems	5
iii)	Write the steps to determine the stability of the characteristic equation using R-H criterion	5
iv)	State advantages and disadvantages of transfer function representation	5
v)	State properties of state transition matrix	5
vi)	State general rules for drawing Root locus of a two real pole system at $s_1 = -2$ and $s_2 = -4$	5
Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Given the control system in figure below find the value of K so that there is 10% error in the steady state.	5
ii)	Find the output response, $c(t)$, for each of the systems shown in figure below. Also find the time constant, rise time, and settling time for each case.	5
OR		
Q2 A	Simplify the following block diagram and obtain transfer function $C(S)/R(S)$	10



Q 2 B	Solve any One	10
i)	Simplify the following Signal flow graph using Mason's gain formula and obtain transfer function $C(S)/R(S)$	10
ii)	Prove that the transfer function of rotational system is given $\frac{\Theta(s)}{T(s)} = \frac{1}{Js^2 + Bs + K}$ Find J and K in the rotational system shown in to yield a 30% overshoot and a settling time of 3 seconds for a step input in torque	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Draw the Bode plot for the following system $G(S) = 100(S+3)/(S(S+1)(S+2))$	10
ii)	Sketch Nyquist plot for the system $G(S) = (S+2)/S^2$	10
iii)	Draw the Polar plot for following system $G(j\omega) = 10/((S+1)(S+3)(S+10))$	10
Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Find the number of poles in the left half-plane, the right half-plane, and on the $j\omega$ -axis for the system shown below 	10
ii)	Given a unity feedback system that has the forward transfer function $G(S) = K(S+4)/(S^2-4S+13)$ do the following: a. Sketch the root locus. b. Find the imaginary-axis crossing. c. Find the gain, K, at the $j\omega$ -axis crossing. d. Find the break-in point. e. Find the angle of departure from the complex pole	10
iii)	Obtain the state variable representation for the given the system transfer function $T(S) = Y(S)/U(S) = 10(S^2+3S+2)/(S^3+3S^2+2S+1)$ where U(s) is the input and Y(s) is the output.	10
Que. No.	Question	Max. Marks
Q5	Solve any four (Short notes / Short question type)	20
i)	Breakaway point in Root locus	5
ii)	Special cases in Routh Hurwitz criterion	5
iii)	Steps to get gain margin and phase margin from Bode plot	5
iv)	Write a short note on state variable representation	5
v)	Conditions for a system to be observable and controllable	5
vi)	Draw the Polar plot for a pole at $s = -a$	5